Ro Min Swe

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PROJECTS

Smart Greenhouse Monitoring System — **Academic Project**
IoT Engineer — *Jan 2026*

- Designed and implemented an ESP32-based IoT monitoring system in Wokwi to simulate real-world greenhouse environmental control scenarios.
- Integrated multiple sensors (DHT22 temperature & humidity, soil moisture, and gas sensor) to collect real-time environmental data.
- Engineered threshold-driven decision logic to trigger alerts and actuator responses (LED and buzzer) based on environmental conditions.

Postgraduate Progress Tracking System — **Full-stack Web Application**
Software Engineer — *Nov 2025*

- Assembled a modular React frontend to support role-based dashboards for Admin, Supervisor, and Student, enabling structured navigation, progress visualization, and file-submission workflows.
- Engineered backend APIs using Express.js to manage authentication, submission tracking, supervisor–student assignments, and version-controlled document workflows.
- Structured and optimized the MySQL database schema to handle multi-role access, file metadata, supervision history, and status transitions.

Handwriting Writer Identification System — **Academic Project**
Deep Learning Engineer — *Nov 2025*

- Developed a CPU-optimized CNN-based writer identification system in Keras, classifying handwritten document images into 70 writers with high accuracy and strong per-class F1 performance on a GPU-free Windows deployment environment.
- Implemented sliding-window image slicing with variance filtering, data augmentation, and label smoothing to improve generalization on variable-length handwriting samples.
- Achieved strong multi-class performance validated using accuracy, confusion matrix analysis, and per-writer F1-score evaluation.

Seizure Forecasting System Using EEG Signals — **Research Project**
Machine Learning Engineer — *Oct – Nov 2025*

- Developed a seizure detection system on CHB-MIT EEG data (~700k windows, 24 subjects) using autoencoder-based anomaly detection combined with XGBoost classification for highly imbalanced data.
- Evaluated performance using accuracy, F1-score, sensitivity, and false-alarm rates, achieving ~99% accuracy while managing severe class imbalance.

EXPERIENCE

Vitrox, — **Penang, Malaysia**
Industrial Trainee — *Mar – Present 2025*

- Developed internal software tools that automated manual workflows for the Vision Inspection team, reducing repetitive processing time by 80% through workflow automation and system integration.
- Collaborated with vision inspection engineers and cross-functional teams to gather requirements and deliver software solutions that streamlined daily operations.
- Enhanced internal applications by implementing new functionalities, fixing bugs, and optimizing system performance, improving usability and reliability.

EDUCATION

Albukhary International University, — **Kedah, Malaysia**
Bachelor of Computer Science — *Expected Graduation: Nov 2026*

- **Concentration:** Data Science, Deep Learning, Natural Language Processing
- **Related Coursework:** Computing in Python, Object-Oriented Programming, Statistics and Probability, Discrete Mathematics, Artificial Intelligence, Information Visualization, Parallel Distributed Database

SKILLS

Programming: Python, JavaScript, SQL, HTML/CSS, C++
Technical Skills: Data Structures & Algorithms, Data Mining & Analytics, Machine Learning
Tools/Platforms: Linux (Ubuntu), Windows, Jupyter, Power BI, Azure, LucidChart, Cisco Packet Tracer